TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE [x]

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

No.	Full name	Phone	**E-Mail**	**Title**
Supervisor	Huỳnh Công Việt Ngữ		nguhcv@fpt.edu.vn	Giảng viên

2. Register information for students (if have)

No.	**Full name**	**Student code**	**Phone**	**E-mail**	**Role in Group**
1	Lê Hồng Đức	SE182106	0928960712	duclhse182106@fpt.edu.vn	Leader
2	Huỳnh Khả Tú	SE182134	0975589955	tuhkse182134@fpt.edu.vn	Member
3	Nguyễn Minh Huy	SE181524	0932556236	huynmse181524@fpt.edu.vn	Member
4	Nguyễn Thị Diệu Vi	SE181505	0948405967	vintdse181505@fpt.edu.vn	Member
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3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

3.1.1. English: ***SME-FraudGuard: A Multi-tenant SaaS Platform for Transaction Anomaly Detection in SMEs***

3.1.2. Vietnamese: ***SME-FraudGuard: Nền tảng SaaS đa tenant hỗ trợ phát hiện bất thường trong giao dịch dành cho doanh nghiệp vừa và nhỏ***

(*) 3.2. Main proposal content (including result and product)

a) Context:

Digital transformation is rapidly increasing the volume of transactions and interactions generated through websites, applications, e-commerce platforms, sales systems, COD orders, and digital services. For small and medium-sized enterprises (SMEs), transaction data is often produced from multiple sources with heterogeneous structures, while the resources required to build a dedicated system for anomaly detection, alert monitoring, and risk-handling support remain limited.

In practice, one group of anomalies can be identified through business conditions that enterprises already know in advance. These cases are well suited to Rule-Based Detection because the conditions are deterministic, controllable, and easy to explain.

However, more complex anomalous behaviors may involve transaction history, activity frequency, changes in behavior over time, or relationships among multiple data features. Such cases are difficult to cover comprehensively using fixed conditions alone. AI therefore becomes an important component for analyzing complex behavioral patterns, identifying potential anomalies, and providing additional information to support enterprise assessment and investigation.

SME-FraudGuard is proposed as a SaaS platform that helps enterprises ingest and normalize transaction data, detect anomalies through Rule-Based Detection or AI-Based Anomaly Detection, generate alerts, support investigation for complex cases, and provide data for monitoring, reporting, and model improvement.

FraudGuard acts as a detection and analysis support layer. Final business decisions remain with the enterprise; the platform does not automatically lock accounts, cancel transactions, reject orders, or directly execute actions in the customer's operational systems.

b) Proposed Solutions:

SME-FraudGuard is designed as a multi-tenant SaaS platform in which each enterprise is provided with an independent workspace to ensure separation of data and operations across customers. Within the project scope, each workspace corresponds to one primary deployment scope and one detection configuration set for that enterprise.

The implementation process begins when the enterprise provides business requirements, data descriptions, required data fields, known anomaly cases, and sample data. The Platform Admin manages customer information, provisions the workspace, and assigns an Implementation Staff member to handle onboarding.

Implementation Staff works with the enterprise to confirm business specifications, input data structures, required fields, and suitable detection conditions. The configuration is tested with sample data before moving to real-data integration.

FraudGuard supports three primary data-ingestion methods: REST API with API Key, Webhook, and CSV/Excel. REST API + API Key is suitable for enterprise systems that actively send transaction data; Webhook supports transaction events sent from customer systems or platforms; CSV/Excel supports batch data, historical data, testing data, or scenarios where real-time integration is not yet required.

After ingestion, data is validated and normalized before being routed to the appropriate detection mechanism.

Free Plan – Rule-Based Detection

For the Free Plan, the Rule Engine is the primary detection mechanism. The rule set is created from business conditions confirmed during implementation, and each transaction is evaluated against the active rules in the workspace.

When a transaction satisfies an anomalous condition, the system creates a Rule Alert and stores the related detection reason for enterprise monitoring. Rule Alerts focus on fast, transparent, and explainable detection for cases with clearly defined business conditions.

Subscription Plan – AI-Based Anomaly Detection

For the Subscription Plan, AI is the primary detection mechanism. The AI Service analyzes transaction data together with related features and historical information to identify anomalous behavioral patterns that are more complex than fixed business conditions.

The AI Service classifies each transaction as Normal or Anomaly. Transactions classified as Anomaly generate AI Alerts for enterprise review and investigation. Related transaction data and historical information are available to support investigation.

AI Alerts can enter an investigation workflow that allows the enterprise to examine transaction context, record conclusions, and build a history of handling complex cases. Investigation results are retained for future reference. Confirmed conclusions can also be aggregated as feedback data for evaluating, refining, and improving subsequent AI model versions.

If the AI Service becomes unavailable, the Rule Engine can be used as a fallback mechanism to maintain the platform's basic detection capability.

Alert Management

FraudGuard manages two alert flows aligned with the characteristics of each detection method. Rule Alerts are generated from preconfigured business conditions and support anomaly notification, display of detection reasons, and historical alert lookup.

AI Alerts are generated from AI analysis of complex data and behavioral patterns. In addition to alerting, AI Alerts support investigation, assignment, conclusion recording, reporting, and investigation history. Confirmed outcomes can become a feedback source for AI evaluation and improvement.

As a result, the Subscription Plan not only expands detection capability but also provides a deeper operational workflow for cases that require business analysis.

Human Oversight

FraudGuard supports enterprises in detection, alerting, analysis, and investigation, but it does not replace customer business decisions. Actions such as accepting or rejecting a transaction, locking an account, canceling an order, or applying other business measures remain the responsibility of the enterprise according to its internal policies and procedures.

c) Functional Requirements:

Module 1 — Authentication, Authorization & SaaS Management

- Support user authentication, session management, and role-based access control for five primary actors: Platform Admin, Implementation Staff, SME Admin, Operator, and Investigator.
- Platform Admin manages customer information, workspaces, and Implementation Staff assignment; Implementation Staff handles onboarding and initial configuration.
- SME Admin manages users and access permissions within the enterprise workspace; Operator monitors detection activities and manages AI Alerts; Investigator performs investigations for assigned cases.
- Ensure users can access only the data and functions permitted by their role and workspace.

Module 2 — Specification Onboarding & Transaction Ingestion

- Support the onboarding process from requirement intake until the workspace is ready for handover to the enterprise.
- Allow Implementation Staff to capture and confirm business requirements, data specifications, required fields, detection conditions, and sample data for testing.

- Support JSON and CSV/Excel data configuration based on confirmed specifications and allow sample-data testing before real-data integration.
- Ingest data through REST API + API Key, Webhook, and CSV/Excel; validate and normalize data before detection.

Module 3 — Rule-Based Detection

- Manage and execute anomaly-detection conditions configured for the enterprise.
- Evaluate each transaction against the active rule set in the workspace; when one or more anomalous conditions are detected, create a Rule Alert and record the triggered rule information.
- Use the Rule Engine as the primary detection mechanism in the Free Plan and as a fallback in the Subscription Plan when the AI Service is unavailable.

Module 4 — AI-Based Anomaly Detection & Fallback

- Allow the AI Service to receive normalized data and related features for anomaly detection.
- Provide a detection result of Normal or Anomaly for each transaction, together with the model/service version for traceability.
- Create an AI Alert when a transaction is classified as Anomaly, supporting enterprise monitoring and investigation.
- Switch to the Rule Engine as a fallback when the AI Service is unavailable; record the detection source for traceability.

Module 5 — Alert, Investigation & Feedback Management

- Manage Rule Alerts and AI Alerts through workflows appropriate to each detection type.
- For Rule Alerts, allow users to view transaction information, triggered conditions, detection time, and alert history.
- For AI Alerts, allow Operators to review alerts, transaction details, and related historical data; perform investigations, record outcomes, prepare reports, or assign an Investigator for deeper analysis.
- Allow Investigators to receive assigned cases, review related information, conduct investigation, record findings, and prepare investigation reports.
- Retain enterprise-confirmed outcomes as investigation history for future reference and as a potential feedback source for AI evaluation and improvement.

Module 6 — Dashboard, Reporting, Notification & Monitoring

- Provide dashboard views covering transactions, detection status, Rule Alerts, AI Alerts, investigations, and detection trends over time.
- Support role-appropriate filtering and reporting for monitoring, operations, and investigation.
- Support suitable notification channels such as in-app messages or email, and allow Platform Admins to monitor overall platform status and major errors.

d) Non-Functional Requirements:

1. Security & Authorization

The system must protect login information, API Keys, and sensitive data. All access requests must be authenticated and authorized according to user roles. API and Webhook ingestion must use appropriate authentication mechanisms to reduce unauthorized access, and sensitive operations must be restricted to authorized actors.

2. Tenant Isolation & Reliability

Each enterprise must have an isolated data scope, and users from one enterprise must not be able to access another enterprise's data. The system should maintain stable data ingestion and processing, limiting transaction loss, invalid data, or uncontrolled duplication that could affect detection results. If the AI Service is temporarily unavailable, the Rule Engine can help maintain basic detection capability for the Subscription Plan.

3. Usability

The interface should be suitable for enterprise users and minimize unnecessary technical complexity. Transaction information, detection results, triggered rule conditions, and investigation results should be presented clearly and understandably according to each actor's responsibilities. Frequently used workflows such as monitoring, alert review, investigation, and reporting should be organized into intuitive interaction flows.

e) Theory & Practical:

- Theoretical foundations: Software-as-a-Service, multi-tenant architecture, role-based access control, transaction data ingestion, data validation and normalization, rule-based anomaly detection, AI-based anomaly detection, human-in-the-loop, alert and investigation management, and feedback-based AI improvement.
- Software architecture: A multi-tenant SaaS platform with major components including SaaS Management & Authorization, Specification Onboarding, Transaction Ingestion, Rule Engine, AI Service, Alert Management, Investigation & Feedback, Dashboard, Reporting, and Notification.
- Processing architecture: Customer Data Sources → Transaction Ingestion → Validation & Normalization → Detection. Rule-Based Flow: Rule Engine → Rule Alert → Monitoring/Notification. AI-Based Flow: AI Service → AI Alert → Investigation → Enterprise Conclusion → Report & Feedback.
- Experimental AI: The project will adapt and evaluate an existing AI model for transaction anomaly detection, providing Normal or Anomaly results. Experiments will assess its suitability for transaction data and detection performance.
- Deployment: The system can run locally or in a demonstration environment using containers; the detailed technology stack, deployment architecture, and quantitative targets will be finalized during the System Design and SRS phases.

f) Products (Expected Deliverables):

- 1. Web Admin Portal & SME Portal: Interfaces for Platform Admin, Implementation Staff, SME Admin, Operator, and Investigator; support workspace and member management, onboarding, monitoring, alerts, investigation, and reporting according to each actor's permissions.
- 2. Transaction Ingestion Platform: Support REST API + API Key, Webhook, and CSV/Excel ingestion, with validation and normalization before data is routed to the detection mechanisms.
- 3. Rule-Based Detection Engine: Provide anomaly detection based on enterprise-configured rules and create Rule Alerts with detection reasons when anomalous conditions are identified.
- 4. AI-Based Anomaly Detection Service: Analyze transaction data to classify each transaction as Normal or Anomaly and provide results for alert generation and enterprise review.
- 5. Alert, Investigation & Feedback Management: Manage Rule Alerts and AI Alerts through their corresponding workflows; support investigation, assignment, findings, conclusion, and

reporting for AI Alerts; retain investigation outcomes for historical reference and as a feedback source for AI evaluation and improvement.

- 6. Dashboard, Reporting & Notification: Provide dashboards for transactions, detection results, alerts, and investigations; support reporting, data export, and suitable notification channels.
- 7. Demo Data, Source Code & Documentation: Provide demonstration data, complete source code, and project documents including SRS, Architecture Design, API Specification, Database Design, AI Design, Test Plan, Deployment Guide, and User Manual.

g) Proposed Tasks:

- Lê Hồng Đức — Project Leader / Backend Developer / Database / Documentation
- Huỳnh Khả Tú — Frontend Developer / AI Engineer /Tester / Documentation
- Nguyễn Minh Huy — Frontend Developer / Tester / Documentation
- Nguyễn Thị Diệu Vi — Backend Developer / Database / Tester / Documentation

4. Other comments (propose all relative things if have):

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HCM, date 14/09/2026

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*

Registration Form – Record of Changes

Form BM03. Attach to the proposed final registration form. The approved project code and title remain unchanged. Export to PDF together with the registration form. File name: SWP490_<ProjectCode>_R2_Registration.pdf

1. Project Information

Project code	FA26SE215
Project name	SME-FraudGuard: A Multi-tenant SaaS Platform for Transaction Anomaly Detection in SMEs
Team	GFA26SE97 — Lê Hồng Đức (SE182106), Huỳnh Khả Tú (SE182134), Nguyễn Minh Huy (SE181524), Nguyễn Thị Diệu Vi (SE181505)
Supervisor	Huỳnh Công Việt Ngữ
Current registration version	Phieu_FA26SE215_v1.docx
Proposed version	Phieu_FA26SE215_v2.docx
Version	1.0 — Record of Changes
Date	02/11/2026

2. Changes

☐No changes to the current registration form

No.	Section	Current content	Proposed content	Reason	Proposed by
1	3.2(a–b) Context and scope	Transaction anomaly detection with risk scores for review prioritization; each enterprise has a workspace and projects, with project-specific detection thresholds.	Clarify known business anomalies versus complex behavioral anomalies. Use one workspace with one primary deployment scope and detection configuration set per enterprise. Retain enterprise control over final business actions.	Focus implementation on workspace-level detection while retaining multi-tenant isolation and human oversight.	Project team
2	3.2(b), (c) Modules 1–2 Roles and onboarding	Explicit RBAC requirements cover Platform Admin and SME Admin. Onboarding uses specification-confirmation statuses; operations staff are described without separate primary roles. SaaS management lists projects, plans, quotas and subscriptions.	Define five primary actors: Platform Admin, Implementation Staff, SME Admin, Operator and Investigator. Add staff assignment, business/data specification confirmation, sample-data testing and handover. Detailed project, quota and subscription-management requirements are no longer explicitly listed.	Make responsibilities and delivery ownership explicit; align access control with implementation, monitoring and investigation workflows.	Project team

No.	Section	Current content	Proposed content	Reason	Proposed by
3	3.2(b), (c) Module 2 Data ingestion	REST API and CSV/Excel ingestion; project API-key lifecycle, detailed validation, custom-field confirmation, quotas and idempotency are specified.	Add incoming Webhook ingestion alongside REST API + API Key and CSV/Excel. Retain validation, normalization and sample testing; describe these at a higher level. Detailed API-key lifecycle, field-change statuses, quotas and idempotency mechanisms are not explicitly restated.	Accommodate event-based customer integrations and align ingestion with confirmed business/data specifications; detailed controls require SRS definition.	Project team
4	3.2(b), (c) Module 3 Rule detection	Dynamic rules include operators, AND/OR groups, risk points, enabled status and SME Admin parameter updates with change history. Triggered rules feed the scoring workflow.	Use active enterprise rules to detect anomalies and create Rule Alerts with triggered conditions/reasons. Keep rules as the Free Plan mechanism and Subscription Plan fallback. Risk points and detailed operator/edit-history requirements are no longer explicitly specified.	Provide transparent detection for known conditions without requiring a normalized risk-scoring pipeline; retain basic detection during AI unavailability.	Project team
5	3.2(b), (c) Old Modules 4–5; new Module 4 AI output and gateway	AI returns risk score, confidence, risk level and explanation/evidence. A separate Risk Score Gateway normalizes scores to 0–100, assigns Low/Medium/High/Critical levels and applies alert thresholds.	AI provides Normal or Anomaly per transaction; Anomaly generates an AI Alert. Retain model/service version and detection-source traceability, plus rule fallback on AI unavailability. Remove the separate Risk Score Gateway and required score, confidence, risk-level and AI-explanation outputs.	Align the AI interface with the intended minimum detection output. Remove unsupported scoring/explanation commitments while retaining alerting, investigation, traceability and fallback; the business workflow remains in scope.	Project team
6	3.2(b), (c) New Module 5 Alerts, investigation and feedback	A common alert/case workflow uses triggered rules or AI explanations, case ownership, notes and named statuses such as Open, Confirmed Fraud and False Alarm.	Separate Rule Alerts for monitoring and reason/history lookup from AI Alerts supporting investigation. Operators may investigate or assign Investigators; retain findings, conclusions, reports and history. Confirmed outcomes may support subsequent AI evaluation/improvement. Exact case statuses are no longer enumerated.	Match workflows to detection type and preserve substantive enterprise review even with binary AI output; capture confirmed outcomes for future evaluation.	Project team
7	3.2(c–d) Dashboard, notifications and quality	Dashboards emphasize risk distribution, risk-level/project filters and scoring sources. Notifications include webhooks/callbacks; CSV/PDF reports, audit trails, privacy masking, health checks and independent scaling are explicit.	Dashboards emphasize transactions, detection status, Rule/AI Alerts, investigations and trends. Keep reporting, export, in-app/email notification and platform monitoring. Consolidate NFRs into security/authorization, tenant isolation/reliability and usability. Detailed export formats, outbound callbacks, masking, audit trails, health checks and scaling requirements are not explicitly restated.	Align presentation with detection/investigation instead of risk scoring. Record reduced specification detail transparently; retained quality goals and implementation controls need elaboration in the SRS.	Project team

No.	Section	Current content	Proposed content	Reason	Proposed by
8	3.2(e–f) Architecture, experimental AI and deliverables	Architecture includes a scoring gateway, workers/queues and audit trails. AI model is unspecified and selection uses demo/synthetic/public data with standardized scoring outputs. Six deliverables combine detection and alert/reporting components.	Architecture explicitly includes onboarding, separate rule/AI detection and investigation/feedback flows. Adapt and evaluate an existing AI model for transaction suitability and detection performance, returning Normal/Anomaly. Reorganize into seven deliverables, separating Rule Engine, AI detection, investigation/feedback and dashboard/reporting. Source code and documentation remain included.	Reuse the existing model subject to validation on transaction data and make component boundaries and deliverables consistent with the revised functional scope.	Project team
9	3.2(g) Team roles	Huỳnh Khả Tú: Frontend Developer / Tester / Documentation. Nguyễn Thị Diệu Vi: Database / Tester / Documentation.	Add AI Engineer to Huỳnh Khả Tú and Backend Developer to Nguyễn Thị Diệu Vi. Other listed responsibilities and team membership remain unchanged.	Assign ownership for AI adaptation/integration and backend implementation in the revised scope.	Project team

Accepted reasons: Review 1 feedback, findings from Iteration 1, or constraints from data, partners or technology. Changes that significantly reduce business complexity must be fully justified.

3. Decision at Review 2 (completed by reviewers)

No.	Decision (Accepted / Adjusted / Rejected)	Reviewer note
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